

Trainings ▾

Initial Check

Misfire Using Heat Indicator Marker

⚠ WARNING ⚠

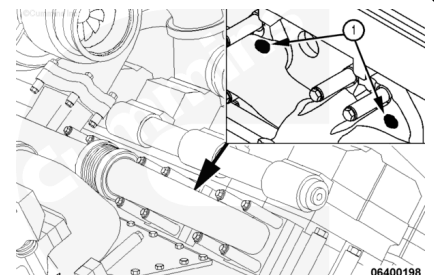
The engine temperature, when performing in-chassis exhaust temperature testing, can be excessive. To reduce the possibility of serious injury wait until the engine coolant temperature is below 50°C [122°F].

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Note : This procedure may **not** be effective when the symptoms occur **only** at low idle or when the engine is cold.

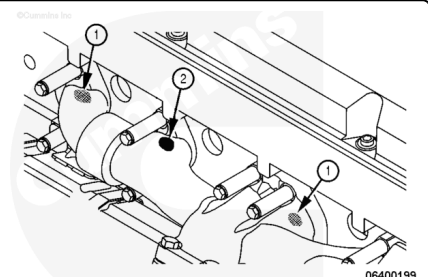
If the engine exhibits symptoms of a possible injector **not** firing, perform the following check:

- Operate the engine to normal operating coolant temperature of 80°C [180°F].
- Stop the engine and allow the exhaust manifold to cool for 10 minutes.
- Apply a mark (1) using a 204°C [400°F] temperature indicator marker, Part Number 3165163, on the surface of all exhaust manifold ports. The mark should be as close to the cylinder head as possible, but **not** directly on the exhaust manifold mounting flange.
- Start the engine and operate under light load for a short time. Do **not** operate the engine under heavy load or for an extended period of time as this may cause inaccurate results.



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If the 204°C [400°F] mark melts (1) the color will disappear and leave **only** a transparent glazed appearance where the mark was. This indicates that the cylinder is operating to a temperature above 204°C [400°F].



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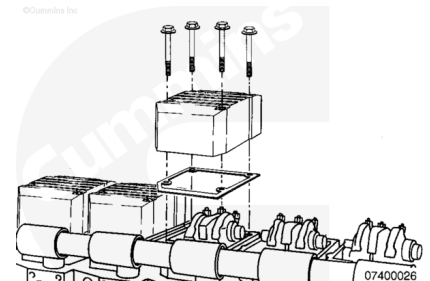
If the 204°C [400°F] mark does **not** melt (2) the mark color remains present. This indicates that the cylinder is **not** operating to temperature of 204°C [400°F], indicating a possible injector malfunction. For additional troubleshooting information, go to Engine Runs Rough or Misfires troubleshooting symptom tree in Section TS.

Note : It is **not** recommended to use the temperature indicator markers to troubleshoot for cylinders that are operating too hot.

Check the injectors (mechanically actuated) when it is suspected they were run dry (without fuel).

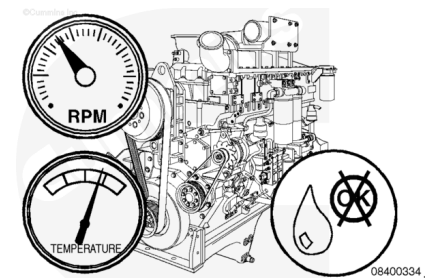
Remove the rocker lever covers and check to be sure all of the injector links are intact. If they are intact, install the rocker lever covers. If they are **not** intact, follow the injector inspection step in this procedure.

Warm up the engine by running at high idle to stabilize the exhaust temperature. If the exhaust temperatures are within 38°C [100°F], then all injectors are firing and working correctly.

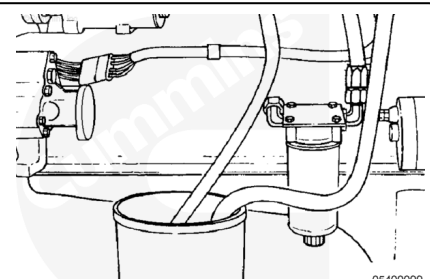


Flourescent Tracer Dye Test

Run the engine until the water temperature is above 80°C [180°F].



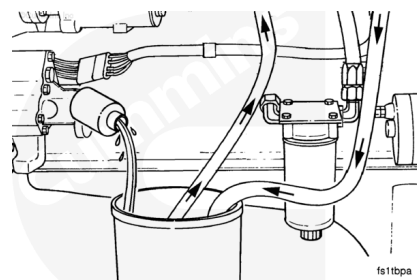
Remove the fuel pump return and supply lines and install a 50 litre [15 gallon] isolated fuel supply tank.



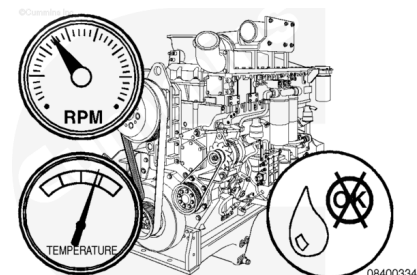
Add one unit of fluorescent tracer, Part Number 3376891, to each 38 litres [10 gallons] of fuel.

Preheat the black light.

Remove the rocker lever cover mounting hardware, but do **not** remove the rocker lever covers.



Idle the engine for 5 to 10 minutes or until normal operating temperature is reached to allow the dye to circulate through the fuel system.

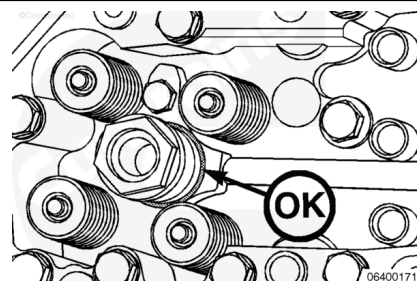


Shut the engine OFF.

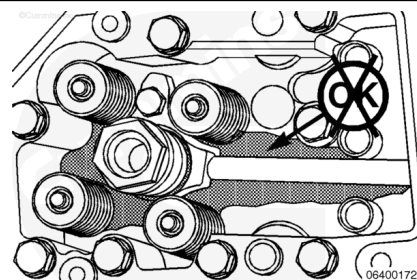
Immediately remove the rocker lever cover.

Use the black light to inspect each cylinder head.

Some leakage around the injector is normal.



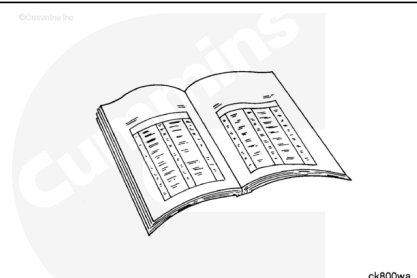
If an abnormal amount of leakage is detected, replace the injector or injector o-rings.



Preparatory Steps

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe



personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-011.html)
- Remove the rocker lever assembly. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-009.html)

Remove

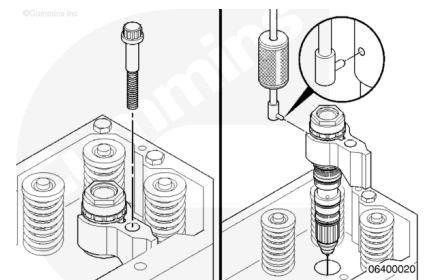
Note : The hold-down clamp can not be removed until the injector is removed.

Remove the injector hold-down capscrew. Remove the injector and clamp.

Use injector removal/installation tool, Part Number 3824830, to remove the injectors. Insert the pin of the tool in the hole provided in the body of the injector. The hole faces the front of the engine.

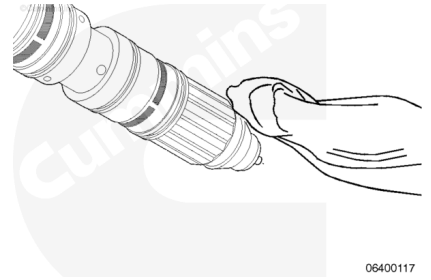
Note : If the injector puller is not available, carefully use a prybar. Pry upward on the injector against the cylinder head.

Cover the injector bore to prevent foreign objects from entering the cylinder.



Clean and Inspect for Reuse

Use a lint-free cloth to clean the exterior of the injector.

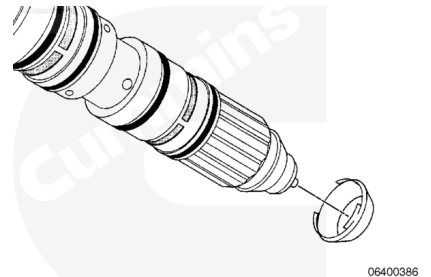


Remove the sealing ring. Note cylinder location and thickness.

Discard sealing ring

Note : Oversize sealing rings are available to adjust injector protrusion in resurfaced and Recon® cylinder heads. Refer to Procedure 002-022 in Section 2. (</qs3/pubsys2/xml/en/procedures/89/89-002-022.html>)

Note : Recon® cylinder heads are marked to identify which sealing ring should be used. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/89/89-002-004-tr.html>)



New o-rings **must** be installed in the correct order on the injector. Take note of the o-ring color and orientation prior to removal.

Remove o-rings 1, 2, 3, and 4.

Use a lint-free cloth. Clean the exterior of the injector.

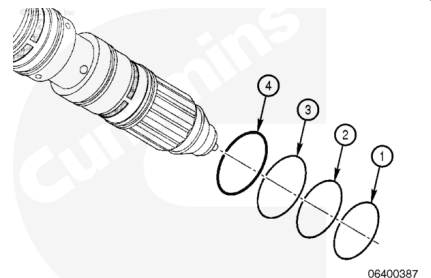
Carefully check the area where the sealing ring touches the injector.

Inspect o-ring grooves for damage.

Inspect injector body and cup retainer for cracks or other damage.

Inspect injector links for excessive wear, pitting or scoring on the ball ends, and other damage.

Note : If the link is damaged or pitting and scoring can be seen or felt, the link **must** be replaced.



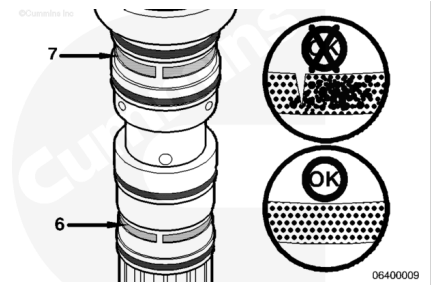
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the

possibility of personal injury.

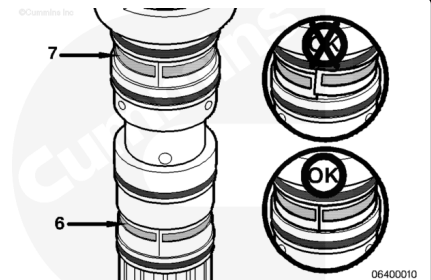
Use solvent when cleaning the filters. The screens **must** be thoroughly dry prior to installation.

Check the rail (6) and timing (7) filter screens for debris, tears, or punctures. Replace the filter screens if damaged or dirty.

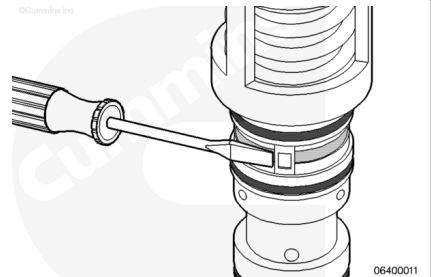


Check the rail (6) and timing (7) filter screens for proper installation.

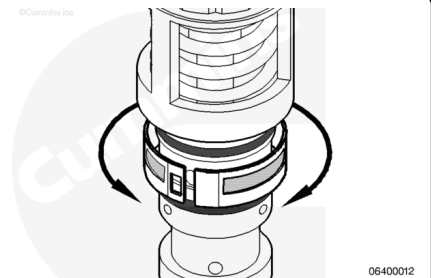
The filter screens **must** be securely snapped into the correct position.



Use a small screwdriver to remove the filter screen by gently prying up on the connector clip.

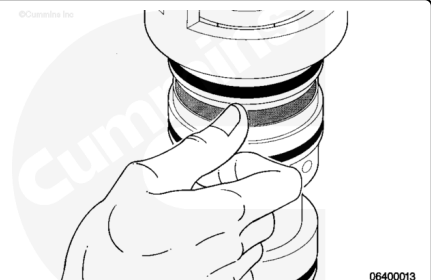


Wrap the filter screen around the inlet groove.



A properly secured connector will create a single snapping sound.

Apply pressure on the filter screen connector clip.



Install

Do **not** lubricate the o-rings until the injector is ready for installation in the cylinder head.

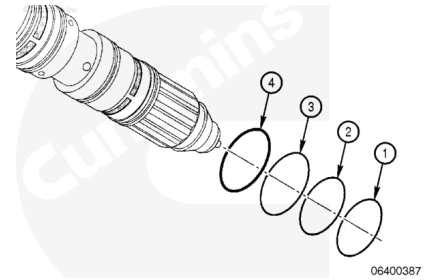
Install four new o-rings on the injector. Do **not** twist the o-rings.

O-ring (1) is brown and is smaller in diameter than the other three.

O-ring (2) is black.

O-ring (3) is brown and is larger in diameter than (1), but has a smaller cross section than o-ring (4).

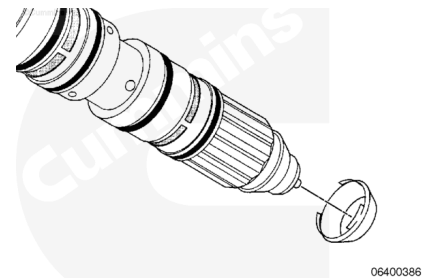
O-ring (4) is brown and is the largest in diameter and cross section.



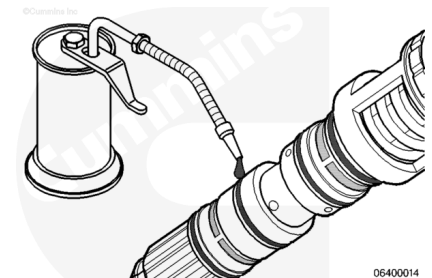
Install a new, proper thickness sealing ring which corresponds to the cylinder location noted during removal.

Note : If installing an injector for the first time into a resurfaced cylinder head, use the following procedure to determine proper sealing ring thickness. Refer to Procedure 002-022 in Section 2.

(/qs3/pubsys2/xml/en/procedures/89/89-002-022.html)



Use clean lubricating engine oil (15W-40) to lubricate the o-rings.



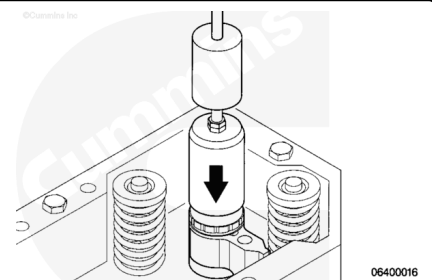
Install the hold-down clamp on the side of the injector.

Install the injector and hold-down clamp in the cylinder head.

Note : If the injector installation tool, Part Number 3824830, is **not** available, use a 40 mm [1 9/16 in] socket with an extension and a rubber mallet to install the injector.

Use injector installation tool, Part Number 3824830, to set the injector in the bore.

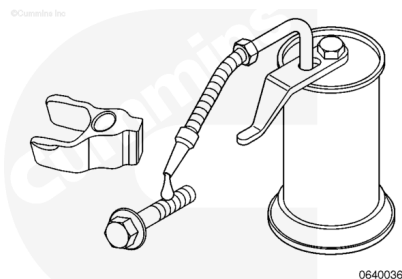
The slide hammer will make a dull sound when the injector is seated properly.



Use clean engine oil to lubricate the injector hold-down capscrew.

Install and tighten the injector hold-down capscrew.

Torque Value:	Step 1	34 N·m [25 ft-lb]
	Step 2	Tighten 90°



Finishing Steps

- Install the rocker lever assembly. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-009.html)
- Adjust the valves and injectors. Refer to Procedure 003-006 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-006.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/89/89-003-011.html)
- Operate the engine and check for leaks.

